

Unit 11: Mechanical Vibrations and Oscillators

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Condensed Cheat Sheet: Springs, Damping, Resonance, and Coupled Motion

Unit 11 at a Glance: Core Sub-Topics

This is the physics payoff of second-order theory. A mass on a spring obeys $mx'' + cx' + kx = F(t)$, and every behavior in the unit — pure oscillation, decay, blow-up, and shared rhythm — is read off that one equation by inspecting its mass, damping, stiffness, and forcing.

- **Simple harmonic motion** — Hooke's law $F = -kx$ and the natural frequency $\omega_0 = \sqrt{k/m}$.
- **Free undamped vibrations** — $x = C \cos(\omega_0 t - \phi)$ and the amplitude-phase form.
- **Damped vibrations** — the discriminant $c^2 - 4mk$ sorts the motion into over-, critically, and underdamped.
- **Forced vibrations and resonance** — driving at ω_0 produces unbounded growth; driving near it produces beats.
- **Coupled oscillators** — eigenvalues of $\mathbf{x}'' = A\mathbf{x}$ give the normal-mode frequencies.
- **The harmonic oscillator** — one equation, $x'' + \omega_0^2 x = 0$, solved four ways and hiding inside almost every stable system.

Part I — Condensed Cheat Sheet

11.1 Simple Harmonic Motion and Hooke's Law

A spring resists displacement with a force proportional to how far it is stretched and directed back toward equilibrium. That single linear law turns Newton's second law into the master oscillator equation.

Hooke's Law and the Natural Frequency

$$F = -kx, \quad mx'' + kx = 0, \quad x'' + \omega_0^2 x = 0, \quad \omega_0 = \sqrt{\frac{k}{m}}.$$

The period and ordinary frequency follow directly from ω_0 :

$$T = \frac{2\pi}{\omega_0}, \quad f = \frac{\omega_0}{2\pi}.$$

A hanging mass settles where gravity balances the spring, $mg = kL$, which is how a measured static stretch L delivers the stiffness $k = mg/L$.

Common Pitfall: Angular Frequency Is Not Frequency

The ω_0 inside $\cos(\omega_0 t)$ is the **angular** frequency in radians per second; the ordinary frequency f in cycles per second is smaller by a factor of 2π , $f = \omega_0/(2\pi)$. A second trap lives one step earlier: after dividing by the mass you read off ω_0^2 , *not* ω_0 . For $2x'' + 18x = 0$ the equation becomes $x'' + 9x = 0$, so $\omega_0^2 = 9$ and $\omega_0 = 3$ — never 9.

11.2 Free Undamped Vibrations

With no damping and no forcing, the imaginary roots $\pm i\omega_0$ give a clean, everlasting sinusoid. Two initial conditions — position and velocity — pin down its two constants.

General Solution and Amplitude-Phase Form

$$x(t) = c_1 \cos(\omega_0 t) + c_2 \sin(\omega_0 t) = C \cos(\omega_0 t - \phi),$$

$$C = \sqrt{c_1^2 + c_2^2}, \quad \tan \phi = \frac{c_2}{c_1}.$$

The amplitude C and phase ϕ depend on the initial data, while ω_0 is fixed by the physical system alone — a stiffer spring (larger k) or a lighter mass raises ω_0 and the motion runs faster.

Solving a Spring-Mass IVP

1. Divide by the mass to reach $x'' + \omega_0^2 x = 0$ and read ω_0 .
2. Write $x = c_1 \cos \omega_0 t + c_2 \sin \omega_0 t$.
3. Apply $x(0)$ to fix c_1 ; differentiate and apply $x'(0)$ to fix c_2 (remember $x'(0) = \omega_0 c_2$).
4. Collapse to $C \cos(\omega_0 t - \phi)$ with $C = \sqrt{c_1^2 + c_2^2}$ to read the amplitude.

11.3 Damped Vibrations and the Damping Discriminant

Add a dashpot and the equation gains a velocity term cx' . The sign of one discriminant decides the entire character of the motion.

The Damping Discriminant

$$mx'' + cx' + kx = 0, \quad \Delta = c^2 - 4mk.$$

Sign	Roots	Regime
$\Delta > 0$	two distinct real	overdamped (no oscillation)
$\Delta = 0$	repeated real	critically damped
$\Delta < 0$	complex $-b \pm i\mu$	underdamped (decaying oscillation)

The critical value $c = 2\sqrt{mk}$ is the boundary; it returns the system to equilibrium fastest *without* overshooting.

Common Pitfall: The Quasi-Frequency Is Not ω_0

An underdamped solution $e^{-bt}(c_1 \cos \mu t + c_2 \sin \mu t)$ oscillates at the **quasi-frequency** $\mu = \sqrt{\omega_0^2 - b^2}$, which is strictly *below* the undamped natural frequency ω_0 . Damping always slows the oscillation; it never speeds it up.

11.4 Forced Vibrations, Beats, and Resonance

Drive the system with $F_0 \cos(\omega t)$ and the steady response depends sharply on how the drive frequency ω compares to the natural frequency ω_0 .

Steady Amplitude, Beats, and Resonance

For undamped forced motion $mx'' + kx = F_0 \cos(\omega t)$ with $\omega \neq \omega_0$,

$$x_p(t) = \frac{F_0}{m(\omega_0^2 - \omega^2)} \cos(\omega t).$$

As $\omega \rightarrow \omega_0$ the denominator vanishes and the amplitude blows up. At exact **resonance** $\omega = \omega_0$ the trial gains a factor of t ,

$$x_p(t) = t(A \cos \omega_0 t + B \sin \omega_0 t),$$

so the amplitude grows linearly without bound. For ω *close to but not equal* to ω_0 the two nearby frequencies interfere to produce **beats** — a fast oscillation under a slow envelope whose period lengthens as $\omega \rightarrow \omega_0$.

Common Pitfall: Resonance Versus Beats

Beats and resonance are not the same phenomenon. **Beats** need $\omega \approx \omega_0$ (close, not equal) and give a *bounded*, slowly modulated amplitude. **Resonance** needs $\omega = \omega_0$ exactly and gives *unbounded* growth from the resonant t factor. Mistaking one for the other misreads whether a structure merely throbs or is being torn apart.

11.5 Coupled Oscillators and Normal Modes

When two masses share a spring, neither equation stands alone: each acceleration depends on the other's position. Writing the system as $\mathbf{x}'' = A\mathbf{x}$ turns the physics into an eigenvalue problem.

Normal Modes from Eigenvalues

$$\mathbf{x}'' = A\mathbf{x}, \quad A\mathbf{v} = \lambda\mathbf{v}, \quad \omega^2 = -\lambda, \quad \omega = \sqrt{-\lambda}.$$

Each eigenvalue gives one **normal mode**: a motion in which all masses oscillate at a single shared frequency with a fixed amplitude ratio (the eigenvector). A system with n degrees of freedom has exactly n modes, and the general motion is a **superposition** of them, each with its own amplitude and phase.

Finding the Normal Modes

1. Assemble the coupling matrix A from the equations of motion.
2. Solve $\det(A - \lambda I) = 0$ for the eigenvalues λ .
3. Convert each to a mode frequency $\omega = \sqrt{-\lambda}$.
4. Find each eigenvector for the mode shape (amplitude ratio), then superpose.

11.6 The Harmonic Oscillator Solved Four Ways

Strip away damping and forcing and one equation remains, $x'' + \omega_0^2 x = 0$. It is worth meeting from several directions because it is the linear heartbeat of nearly every stable system.

One Equation, Many Faces

$$x'' + \omega_0^2 x = 0 \implies r^2 + \omega_0^2 = 0 \implies r = \pm i\omega_0 \implies x = c_1 \cos \omega_0 t + c_2 \sin \omega_0 t.$$

The energy method — multiply by x' and integrate — yields the conserved quantity

$$\frac{1}{2}(x')^2 + \frac{1}{2}\omega_0^2 x^2 = \text{const.}$$

The same equation governs an ideal LC circuit $Lq'' + \frac{1}{C}q = 0$ with $\omega_0 = 1/\sqrt{LC}$, because near any stable equilibrium a smooth potential looks quadratic and its restoring force looks linear.

Unit 11 Key Takeaway

Everything flows from $mx'' + cx' + kx = F(t)$. The **natural frequency** $\omega_0 = \sqrt{k/m}$ sets the rhythm; the **discriminant** $c^2 - 4mk$ sorts damped motion into over-, critically, and under-damped; **resonance at** $\omega = \omega_0$ forces a t factor and unbounded growth while a near-match gives **beats**; and the **eigenvalues of** $\mathbf{x}'' = A\mathbf{x}$ unlock the normal modes of coupled motion. Underneath it all sits the **harmonic oscillator** $x'' + \omega_0^2 x = 0$, the universal linear model of a system near stable equilibrium.